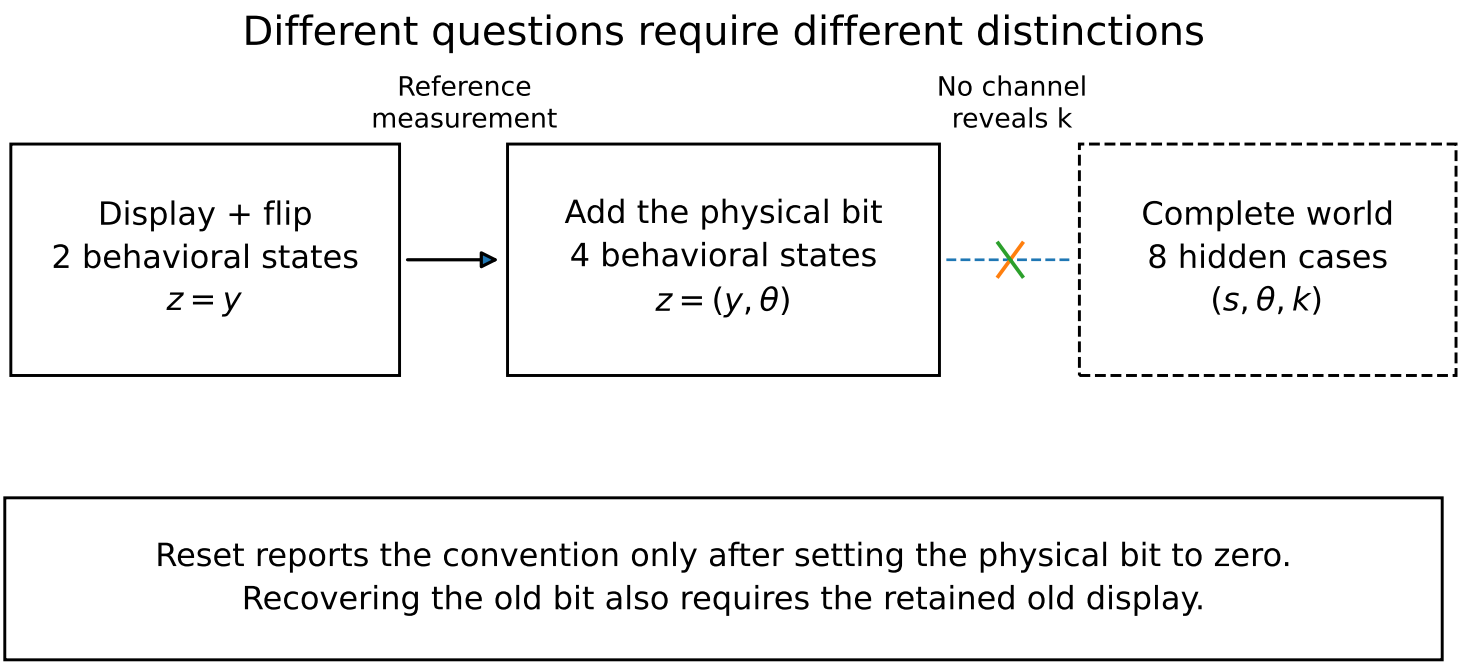
Different questions require different distinctions
Reference measurement
No channel reveals k
Display + flip
2 behavioral states
$z = y$
Add the physical bit
4 behavioral states
$z = (y, \theta)$
Complete world
8 hidden cases
(s, θ, k)
Reset reports the convention only after setting the physical bit to zero.
Recovering the old bit also requires the retained old display.